

Stock Return Distribution, Volatility, Predictability and Efficiency in the BRICS

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Abstract

This paper analyzes the characteristics of stock returns for Brazil, Russia, India, China and South Africa (BRICS) for the period 1995 to 2014. Our main findings are as follows: the distribution of stock returns for the BRICS exhibits peakedness with fatter and longer tails. This is invariant to both the unit of measurement and the time horizon in which returns are studied. Return predictability and volatility clustering is established for all markets. We do not find evidence for the so called leverage effect.

Keywords: Return Predictability, Normality, Leverage effect, Volatility, Efficiency, Emerging Market

1. Introduction

Since Goldman Sachs economist, Jim O'neil coined the term BRIC in the early 2000s, the economies of Brazil, Russia, India and China have taken centre stage in both the global political and economic arena. In 2010, South Africa joined the club, officially spreading the tentacles of the largest emerging market economies over four continents. By 2013, the BRICS accounted for almost 3 billion of the world's population, with a combined nominal GDP of US\$16.039 trillion. About US\$4 trillion of foreign reserves are held by the BRICS, with China alone accounting for more than a quarter.

Over the past few years, the performance of BRICS stock markets has been sterling. Data from Reuters (2012) shows that viewed in a 10 year horizon, the Morgan Stanley Capital International (MSCI) BRIC index returned a striking 450%, compared to the 320% and 98% returns on other

emerging and developed markets respectively. Between 2001 and 2007 the MSCI's BRIC index returned over 500%, significantly outperforming emerging markets. However, recent evidence shows that the hay days may be over soon. There have been slumps in the most recent period with losses of 8.6% in the past five years in dollar terms. There are also indications that China's impressive double digit growth spurt is fading. Brazil and South Africa's growth has been anaemic, and Russia faces problems in the oil and gas sector while reforms in India have been sluggish. The volatility in growth rates and stock market performance raises important questions pertinent to investments, portfolio diversification and the overall role of the BRICS in global economic growth. Will the BRICS assets continue to receive the attention they have enjoyed over the past decade? What is the nature of the risk return relationship in these markets? Questions such as these bother on the distributional patterns, volatility and predictability of stock returns as well as the efficiency of the BRICS markets. This article concerns itself with return distributions, volatility and predictability and their implications for market efficiency in the BRICS. The study seeks to extend the literature in two directions.

The first is methodological. Standard asset pricing models such as the mean variance model takes the normality of asset returns as given. Although this assumption has been pointed out to be highly unrealistic (see Mandelbrot, 1963; Rachev, 2003) even for developed markets returns, a significant amount of research continue to focus on the normality of returns as a starting point. This is not surprising as the computational intensity underlying alternative distributions is both time consuming and more daunting. More so, the properties of the normal distribution are sufficiently well known and studied in the literature. However, the consequence of relying on models of normal returns may lead to significant underestimation of the risk of investing in emerging markets, particularly if the distribution is skewed and fat tailed. This paper thus questions the over reliance on the normality assumption that exist in the extant literature on the distribution of returns in emerging markets. A formal test of skewness and kurtosis using Mardia (1970) skewness and kurtosis test and joint skewness kurtosis test of Henze and Zirkler (1990) are performed on each of the BRICS market index return series to establish evidence on the nature of the return distributions.

The second contribution is to extend the literature on the peculiarities of asset returns in BRICS. A number of research efforts have been expended in understanding the return distribution of emerging markets generally, however, a lot remains to be learned about the BRICS stock markets efficiency in allocating scarce resources. Moreover, the most comprehensive study of the return

distribution of emerging markets appeared nearly two decades ago (Bekaert et al, 1998). On the empirical regularities emerging markets are noted to have low/or negative correlations with the more developed world (see Harvey, 1995; Alagidede, 2010); emerging market economies offer returns that exceed industrial-market returns (Buckberg, 1995). Both of these facts suggest that unexploited profit opportunities may exist. At the same time, emerging market returns tend to exhibit high volatility and autocorrelation, long run predictability and generally low levels of liquidity (Bekaert and Harvey, 1997; Aggarwal et al., 1999; Kasman et al, 2009; Blitz et al., 2013; Hull and McGroarty, 2014). These stylized features may signal market inefficiency and opportunities for profitable arbitrage. Understanding the dynamic behaviour of stock returns in the BRICS is crucial for portfolio managers, policy makers, and researchers. This article extends our understanding of these matters by accounting for return dynamics in different time horizons and currencies.

1.1. Stylized facts of BRICS stock markets

The key facts about BRICS stock markets are indicated in Table 1. For the sake of brevity, and in line with data availability, the World Development Indicators for the stock market variables are only reported for 2012. The market capitalization, turnover ratio and trading value are all expressed as a percentage of economic activity. Market capitalization is the share price multiplied by the number of shares outstanding, and it is a rough benchmark for judging a company's net worth. The turnover ratio is derived by dividing the value of total shares by the market capitalization. While the total value traded ratio captures trading relative to the size of the economy, and the turnover ratio measures trading relative to the size of the stock market. In practice, the turnover ratio proxies the liquidity of the market: high turnover is an indicator of low transaction costs.

Table 1: Facts about BRICS Stock Markets (2012)

	Market Capitalization (Current US\$)	Market Capitalization (% of GDP)	Listed domestic companies	Total value traded (% of GDP)	Turnover ratio (%)
Brazil	\$1.229 trn	54.60	353	37.05	67.88
Russia	\$874 bn	43.41	276	36.34	87.64
India	\$1.26 trn	68.60	5191	33.80	54.63
China	\$3.69 trn	44.94	2494	70.82	164.44
South Africa	\$612 bn	159.33	348	81.13	54.93

Source: Compiled from World Development Indicators of the World Bank (Online version)

From Table 1, no single BRICS country dominates in terms of all indicators, unsurprisingly confirming the diversity of depth, performance and influence of the national stock exchanges. Judged by market capitalization, the Chinese market stands out. The Shenzhen, Shanghai and Hong Kong Stock markets had a market capitalisation of \$3.7 trillion dollars at the end of 2012. The Shenzhen stock exchange is overwhelmingly dominated by state owned enterprises which are the back bone of the Chinese economy, while the Shanghai is not fully opened to foreign investors. Brazil and India have market capitalisation of about \$1.2 trillion as of 2012. In the BRICS, Russia and South Africa are the smallest markets using this indicator at \$874 billion and \$612 billion respectively. In relation to the size of the domestic economy, however, South Africa dominates as seen from Table 1. The size of the stock market as a proportion of GDP is a whopping 159%. This gives a high value of shares traded as proportion of GDP in South Africa (81%) than any country in Table 1. Interestingly, China's stock market is 44% of GDP, slightly bigger than the Russian 43% but less than Brazil's 54% and India's 68%. With the exception of South Africa and China, total value traded as a share of GDP is less than 40% as of 2012.

The number of listed domestic companies amounted to 5191 in India in 2012. This is about 15 times the number of companies in Brazil and South Africa and about 19 times the number of domestic companies listed in Russia's stock market. China comes second with 2494 companies. The most liquid of the BRICS stock markets is China (164%), followed by Russia (87%) and Brazil (67%). India and South Africa have a turnover ratio of about 54%.

2. Empirical Strategy and Data

The analysis of the data for this study follows three important steps. First we examine the nature of the probability distribution of the index return series for the BRICS measured in both US dollars and local currency units and for their different holding periods: daily, weekly and monthly. While this analysis is an end in itself, it also offers important information relevant for selecting the appropriate statistical model for performing inference on the return generating process. To achieve this aim, we employ the Mardia (1970) skewness and kurtosis and Henze and Zirkler (1990) test for joint skewness and kurtosis. These formal tests for normality give much reliable evidence that rely on descriptive statistics as is common in the literature. The second step involve testing for the presence of autoregressive conditional heteroscedasticity (ARCH) effect in the mean of the index returns for each of the BRICS. Again, while the presence of ARCH effect in the mean of the return series offer important information about the

return behaviour and efficiency of markets, it also offers vital information in selecting the appropriate model for the return generating process. As a final step, we estimate a statistical model that appropriately captures the essential features of the return generating process.

The rests of this section is devoted to the description of the econometric or estimation technique adopted for this paper and brief description of our data. To begin with, we assume that the log of the stock price index follows a random walk with a drift. That is:

$$\ln P_t = \mu + \ln P_{t-1} + \varepsilon_t \quad (1)$$

Equation (1) implies the following expression for the return generating process (RGP):

$$R_t = \ln P_t - \ln P_{t-1} = \mu + \varepsilon_t \quad (2)$$

In many applications, it turns out that the effect of positive news tends to differ from negative news of equal size, on average. To account for this, we adopt the exponential generalized autoregressive conditional heteroschedasticity in the mean (EGARCH-M) of the return series. We also augment the standard EGARCH-M specification with $ARMA(p, q)$ model. The specific functional form of the model adopted for this study is:

$$R_t = \mu + \sum_{i=1}^p \beta_i R_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \phi h_t + \varepsilon_t \quad (3)$$

$$\varepsilon_t | \eta_{t-1} \approx t.d(0, h_t, \nu_t) \quad (4)$$

With the above specification, R_t is the market return and μ is the expected return, h_t is the risk premium (volatility), ε_t is the error term that we assume to have a student-T distribution due to the skewness and excess kurtosis of each of the series, while p and q are the optimal lag orders for the autoregressive (AR) and moving average (MA) terms respectively. The inclusion of the risk premium term in the mean equation (4) is to model the relationship between expected return and risk. A significant and positive β_i implies that expected return is positively related to risk and a negative relation if the reverse is true.

Additionally, to account for possible asymmetry of news on return-risk relationship, we apply the exponential generalized autoregressive conditional heteroschedasticity (EGARCH) model proposed by Nelson (1991), in which the volatility (h_t) as presented in equation (3) is express as;

$$\ln(h_t) = w + a_1 z_{t-1} + g_1(|z_{t-1}| - E(|z_{t-1}|)) + d_1 \ln(h_{t-1}) \quad (5)$$

This specification is much more flexible since we do not need to explicitly impose non-negativity restrictions on the parameters in the ARCH and GARCH terms in the model. The reason is that the log transformation ensures that the variance is non-negative and therefore free from the problems of possible negative variance as in the case of the standard GRACH models. The a_1 is the parameter that capture leverage effect, implying that a significant positive a_1 is an indication of positive innovations having more effects on returns than negative innovations, g_1 is the symmetry term and d_1 the GARCH term, while w is the constant term in the volatility equation.

2.1. Data Sources and Descriptions

The data employed in this paper is the Morgan Stanley Capital International (MSCI) indices for emerging markets, focussing mainly on Brazil, Russia, India, China and South Africa. We use the MCSI World Index as a proxy for the World portfolio. The MSCI data for emerging markets captures large and mid-capitalisation stocks and it covers roughly 85% of the free float-adjusted market capitalization in each country. The data is from the period January 1995 to May 2014. For completeness we use daily, weekly and monthly data reported in both US dollars and local currency values. This allows us to compare the behaviour of returns across different time horizons and to examine the impact of exchange rate variations on stock returns. All the series under consideration are obtained from Thomson DataStream. Following Pukthuanthong and Roll (2009), we filter returns to purge holidays and nontrading days.

3. Emerging Market Return Distribution

In this section we examine the distribution of the return series. First we look at the summary statistics from the return series. Next we test for the presence of skewness and kurtosis in the return series (both in US dollars and local currency units) on daily, weekly and monthly frequencies. Summary statistics on all the series that we examined are reported in the appendix. Judging by the Jarque-Bera statistic, the reported summary statistics on the return series indicate non-normal distribution for all the series both in US dollars and local currency for daily, weekly and monthly holding periods. In almost all the cases examined here, the skewness coefficient is negative with the exceptions being daily dollar returns for China and daily returns in local

currency units for Brazil and China. Thus the implication from the summary statistics suggest that the probability distributions of BRICS returns for daily, weekly and monthly holding periods are negatively skewed, indicating that the left tails of the distribution are either longer or fatter (or both) than the right tail.

Table 2: Skewness and Kurtosis Test on Daily Returns

	Brazil	Russia	India	China	South Africa	World
Daily US Dollar Returns						
Skewness	0.0094 [7.946] (0.0048)	0.9947 [168.3] (0.0000)	0.0029 [2.408] (0.1207)	0.0008 [0.697] (0.4038)	0.1759 [148.5] (0.0000)	0.1376 [116.16] (0.0000)
Kurtosis	10.5850 [12127.2] (0.0000)	14.8363 [29531.3] (0.0000)	9.6360 [9282.4] (0.0000)	8.7725 [7023.9] (0.0000)	8.4823 [6335.3] (0.0000)	10.8188 [12886.4] (0.0000)
Joint	81.462 [96.73] (0.0000)	166.91 [126.78] (0.0000)	71.39 [91.64] (0.0000)	114.81 [110.59] (0.0000)	61.2102 [85.881] (0.0000)	91.1044 [101.15] (0.0000)
Daily Local Currency Returns						
Skewness	0.7469 [29.637] (0.0000)	1.2042 [47.779] (0.0000)	0.2847 [11.296] (0.0008)	0.2967 [11.771] (0.0006)	0.5797 [23.0000] (0.0000)	2.2989 [91.214] (0.0000)
Kurtosis	5.7088 [70.931] (0.0000)	7.4317 [189.854] (0.0000)	4.8019 [31.386] (0.0000)	4.1349 12.451 (0.0004)	5.0858 [42.054] (0.0000)	8.4045 [282.352] (0.0000)
Joint	2.5986 [10.005] (0.0016)	3.912 [14.423] (0.0001)	1.6422 [6.008] (0.0142)	3.0689 [11.680] (0.0006)	1.7083 [6.313] (0.0120)	5.6917 [19.178] (0.0000)

Note: In [] are the chi squared statistics and () are the probability values.

Following the clue from the summary statistics, we undertake a formal statistical test to produce evidence on the skewness and excess kurtosis of the return series for the BRICS. The results are reported in Table 2 (for daily returns), Table 3 (for weekly returns) and Table 4 (for monthly returns).

Three different tests are employed – Mardia skewness test, Mardia kurtosis test and Henze-Zirkler joint skewness and kurtosis test. As can be seen from the upper panel of Table 2, the null hypothesis that the skewness coefficient is not statistically different from zero for the daily return series is flatly rejected by the Mardia skewness test for Brazil, Russia, South Africa and the World. In the case of India and China, however, the null hypothesis is not rejected at the conventional level of statistical significance. There is strong evidence of excess kurtosis (leptokurtosis) in the daily stock return series for the BRICS market indexes as well as the World stock market index. The Mardia kurtosis test for excess kurtosis in the return series flatly rejected

the null hypothesis in all the cases at 1% level of statistical significance. The strong evidence of excess kurtosis in the daily dollar return series is confirmed by Henze-Zirkler joint test for the evidence of skewness and excess kurtosis. The null hypothesis is rejected at the 5% level of statistical significance or lower. The evidence reported here gives strong indication that the probability distribution of the daily dollar stock returns for the BRICS exhibits peakedness with fatter and long tails. Jointly, these indicate that the probability distribution is non-normal, and special attention needs to be taken when performing statistical inference on them.

Table 3: Skewness and Kurtosis Test on Weekly Stock Returns

	Brazil	Russia	India	China	South Africa	World
Weekly US Dollar Returns						
Skewness	0.4231 [71.723] (0.0000)	0.0156 [2.654] (0.1033)	0.1361 [23.073] (0.0000)	0.0722 [12.236] (0.0000)	0.0372 [6.302] (0.0121)	1.2880 [218.326] (0.0000)
Kurtosis	7.0750 [699.5] (0.0000)	8.5842 [1313.6] (0.0000)	5.2467 [212.639] (0.0000)	5.9561 [368.1] (0.0000)	7.6655 [916.942] (0.0000)	12.8531 [4089.623] (0.0000)
Joint	11.6893 [31.589] (0.0000)	28.2706 [51.143] (0.0000)	3.9810 [14.085] (0.0002)	10.9712 [30.365] (0.0000)	13.2744 [34.115] (0.0000)	13.4321 [34.355] (0.0000)
Weekly Local Currency Returns						
Skewness	0.29887 [50.627] (0.0000)	0.0108 [1.824] (0.1769)	0.10997 [18.640] (0.0000)	0.0761 [12.891] (0.0003)	0.2023 [3.430] (0.0640)	1.0809 [183.207] (0.0000)
Kurtosis	6.8719 [631.514] (0.0000)	9.1566 [1596.713] (0.0000)	5.2756 [218.146] (0.0000)	5.9893 [376.418] (0.0000)	6.1066 [406.560] (0.0000)	12.2076 [3571.348] (0.0000)
Joint	11.2145 [30.786] (0.0000)	33.7756 [55.649] (0.0000)	5.0106 [17.237] (0.0000)	11.2019 [30.764] (0.0000)	11.4684 [31.218] (0.0000)	12.1046 [32..273] (0.0000)

Note: In [] are the chi squared statistics and () are the probability values.

This conclusion is invariant to the unit in which the returns are measured. This is confirmed by the results reported in the lower panel of Table 2 where the null hypotheses of no skewness, no kurtosis and no joint skewness and kurtosis in daily stock returns in local currency units are all rejected at conventional level of significance. It has been pointed in the extant literature that the normality assumption on the probability distribution of stock returns are more likely to be violated in high frequency data than a low one. We therefore repeat the above skewness and kurtosis test on weekly and monthly stock returns for the BRICS as well as the World. Again, we consider both the returns measured in local currency units and the US dollar. The findings on the weekly returns series concerning skewness and kurtosis of their underlying probability

distributions are not at variance with those reported on daily return series. With the exception of Russia where the null hypothesis of no skewness is not rejected for both returns in the local currency units and US dollar weekly series, the null hypothesis of no skewness in the probability distribution of returns is rejected for the remaining BRICS weekly return series. In terms of kurtosis, the null hypothesis is rejected for all the weekly return series measured in terms of both the US dollar and local currency units.

Table 4: Skewness and Kurtosis Test on Monthly Stock Returns

	Brazil	Russia	India	China	South Africa	World
Monthly US Dollar Returns						
Skewness	1.2993 [51.553] (0.0000)	1.1155 [44.258] (0.0000)	0.4244 [16.837] (0.0000)	0.2909 [11.543] (0.0007)	2.1596 [85.688] (0.0000)	2.8078 [111.408] (0.0000)
Kurtosis	6.9145 [148.126] (0.0000)	7.0115 [155.554] (0.0000)	5.1774 [45.829] (0.0000)	4.1089 [11.887] (0.0006)	10.0213 [476.548] (0.0000)	9.7901 [445.684] (0.0000)
Joint	2.4569 [9.462] (0.0021)	3.5946 [13.443] (0.0002)	1.2176 [3.949] (0.0469)	3.0417 [11.610] (0.0007)	2.2856 [8.785] (0.0030)	5.1941 [17.956] (0.0000)
Monthly Local Currency Returns						
Skewness	1.2993 [51.553] (0.0000)	1.1155 [44.258] (0.0000)	0.4244 [16.837] (0.0000)	0.2909 [11.543] (0.0007)	2.1596 [85.688] (0.0000)	2.8078 [111.408] (0.0000)
Kurtosis	6.9145 [148.126] (0.0000)	7.0115 [155.554] (0.0000)	5.1774 [45.829] (0.0000)	4.1089 [11.887] (0.0006)	10.0213 [476.548] (0.0000)	9.7901 [445.684] (0.0000)
Joint	2.4569 [9.462] (0.0021)	3.5946 [13.443] (0.0002)	1.2176 [3.949] (0.0469)	3.0417 [11.610] (0.0007)	2.2856 [8.785] (0.0030)	5.1941 [17.956] (0.0000)

Note: In [] are the chi squared statistics and () are the probability values.

The Henze-Zirkler joint test for skewness and kurtosis in the probability distribution of weekly BRICS returns (both local currency and US dollar units) rejected the null hypothesis flatly for all the BRICS market as well as the World stock market returns. The implication here is that the weekly return series inherits the fatter and longer tails and peakedness that characterized the probability distribution of daily stock returns, at least for the BRICS and the World stock market as a whole.

We now consider the tail and peak behaviour of the probability distribution of monthly index returns for the BRICS and the World Stock markets. In the preceding paragraph, we arrived at the conclusion that weekly return series inherits the properties of the distribution of daily returns. It is thus important to ask whether monthly returns also inherits the properties of the

probability distribution of weekly returns and hence those of daily returns. We thus search for evidence of skewness and kurtosis in monthly returns for the BRICS in both dollar series and local currency series. The results are reported in Table 4. Consistent with our results on daily and weekly returns, the Mardia tests for skewness and kurtosis rejected the null hypothesis of no skewness and no kurtosis in the probability distribution of monthly stock returns. Henze-Zirkler joint test for skewness and kurtosis in the probability distribution of monthly returns also rejected the null hypothesis in all cases. This conclusion is not sensitive to the unit in which monthly returns are measured (either in US dollars or local currency units). Thus the evidence of non-normality in the distribution of BRICS index and World market index returns is robust. In line with Bekaert et al (1998) we are able to establish that the BRICS stock returns exhibit time varying skewness and kurtosis. The underlying structure of the BRICS: their potential for large growth swings, susceptibility to regulatory and political changes, and continued integration into the global economic and financial system may thus lead their stock returns to deviate significantly from the Gaussian assumption. As a result, any statistical inference which relies on the normality of the distribution of returns has the potential of making erroneous conclusions.

4. Volatility, Predictability and Risk-return relationship

In this section of the paper we estimate and report evidence on the volatility, predictability and risk-return relationships for the BRICS index returns. Due to the similarities in the distribution of the return series for the daily, weekly and monthly holding periods, we only focus on modelling daily return behaviour. To begin with, we test for the presence of ARCH effects in the mean of the daily returns using the Lagrange Multiplier (LM) test. The results are reported in Table 5.

Table 5: Lagrange Multiplier Test for ARCH effects in BRICS Daily Stock Returns

	Brazil	Russia	India	China	South Africa
Lagrange Multiplier Test for ARCH effect in the Mean of Daily US dollar Returns					
Lags =1	222.26 (0.000)	549.76 (0.000)	136.23 (0.000)	361.35 (0.000)	289.60 (0.000)
Lags=5	1079.97 (0.000)	641.59 (0.000)	317.96 (0.000)	736.58 (0.000)	638.81 (0.000)
Lags=10	1167.28 (0.000)	732.62 (0.000)	413.28 (0.000)	802.37 (0.000)	827.59 (0.000)
Lags=20	1235.45 (0.000)	871.65 (0.000)	449.68 (0.000)	910.10 (0.000)	988.58 (0.000)
Lagrange Multiplier Test for ARCH effect in the Mean of Daily Local Currency Returns					
Lags =1	189.52 (0.000)	571.66 (0.000)	183.34 (0.000)	361.33 (0.000)	330.97 (0.000)
Lags=5	563.23 (0.000)	662.92 (0.000)	385.87 (0.000)	739.24 (0.000)	547.31 (0.000)
Lags=10	621.29 (0.000)	753.76 (0.000)	463.03 (0.000)	804.67 (0.000)	589.44 (0.000)
Lags=20	658.83 (0.000)	893.91 (0.000)	498.62 (0.000)	912.78 (0.000)	639.78 (0.000)

Note: The test statistic for the LM test follows a chi square distribution. The numbers reported in the above table are the chi squared statistics with () housing the probability values.

The results of the Lagrange Multiplier (LM) test for the presence of ARCH effects in the mean of BRICS index return series flatly rejected the null hypothesis of no ARCH effects at 1% level of statistical significance for all the countries. This conclusion is invariant to whether the BRICS index returns are measured in the local currencies or in United States dollars. The invariance of the presence of ARCH effects in the mean of the returns for the BRICS suggests that exchange rate movements have not affected BRICS index returns volatility; an empirically testable proposition. We, therefore, conclude that there is significant evidence that the mean of the return series for the BRICS is volatile. The variance is time dependent and thus need to be accounted for in any statistical model aimed at capturing the return generating process of these countries.

Table 6: Estimated ARMA-EGARCH-M Model of Daily Returns in Dollars

	Brazil	Russia	India	China	South A	World
π	0.125	0.248***	0.0125	-0.0727	-0.332	0.0871***
	(1.49)	(4.27)	(0.09)	(-1.32)	(-0.98)	(8.26)
$ARCHM$						
J	0.0237	-0.00848*	0.0722	0.0220*	0.126	-0.0251**
	(1.90)	(-2.57)	(1.85)	(2.11)	(1.21)	(-2.74)
ARMA						
L1.ar	0.999***	-0.0126	0.990***	0.998***	-0.00115	0.0573***
	(2710.22)	(-1.10)	(4905.07)	(633.92)	(-0.08)	(9.54)
L2.ar		0.000151	0.00981***		0.00808	0.0272**
		(0.02)	(48.78)		(0.68)	(2.79)
L3.ar					-0.0111	
					(-0.98)	

L1.ma	-0.997***		-0.997***	-0.997***		-0.0646***
	(-1617.12)		(-1370.94)	(-573.32)		(-10.99)
<i>ARCH</i>						
<i>a</i>	0.0221	0.0520	0.0226	0.0147	0.0229	-0.0135
	(1.10)	(1.26)	(0.67)	(0.44)	(0.77)	(-0.48)
<i>g</i>	0.129***	0.362***	0.117**	0.235***	0.100***	0.156***
	(4.52)	(5.40)	(2.75)	(4.68)	(3.35)	(4.06)
<i>d</i>	0.815***	0.135	0.0166	0.189	-0.00775	-0.267
	(28.66)	(0.77)	(0.05)	(1.50)	(-0.01)	(-1.65)
<i>g + d</i>	0.944	0.497	0.134	0.424	0.092	-0.111
<i>w</i>	0.352***	2.631***	1.209**	1.416***	1.187	0.176
	(6.85)	(4.84)	(3.20)	(6.25)	(0.87)	(1.86)
<i>N</i>	5059	5059	5059	5059	5059	5059
<i>AIC</i>	21709.7	23600.2	19077.8	20080.7	19084.4	13097.5
<i>BIC</i>	21768.4	23658.9	19143.1	20139.5	19149.7	13162.8
<i>LBQ(11)</i>	15.702 [0.152]	6.412 [0.844]	16.463 [0.124]	9.342 [0.590]	17.199 [0.102]	12.288 [0.342]
<i>LBQ(25)</i>	27.412 [0.335]	32.808 [0.135]	33.033 [0.130]	30.928 [0.191]	29.527 [0.242]	28.546 [0.283]
<i>LBQ²(11)</i>	14.460 [0.208]	6.424 [0.843]	16.267 [0.131]	9.204 [0.603]	16.692 [0.117]	11.300 [0.418]
<i>LBQ²(25)</i>	26.749 [0.368]	32.161 [0.153]	32.592 [0.141]	29.291 [0.251]	28.983 [0.264]	27.795 [0.317]

t statistics in parentheses and the numbers in square brackets are the *p*-values, *LBQ* is the Ljung-Box statistics for serial correlation and *LBQ²* is the Ljung-statistics for ARCH effect, with the lags used in the parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 7: Estimated ARMA-EGARCH-M model for Daily Returns in Local Currency

	Brazil	Russia	India	China	South A	World
<i>n</i>	-0.121***	0.197**	0.0537	-0.0837***	0.138	0.103***
	(-10.43)	(2.75)	(0.51)	(-18.55)	(0.94)	(6.53)
<i>ARCHM</i>						
<i>J</i>	0.0460***	-0.00429	0.0603	0.0184***	0.0462	-0.0391**
	(11.57)	(-1.36)	(1.66)	(16.58)	(0.56)	(-2.77)
ARMA						
L1.ar	0.00760	-0.0105	0.993***	-0.987***	1.013***	-0.00778
	(0.63)	(-0.85)	(4511.98)	(-835.39)	(7265.27)	(-0.60)
L2.ar		0.00416	0.00611***		-0.00813***	0.0230*

		(0.45)	(26.74)		(-64.04)	(2.57)
L3.ar					-0.00542***	
					(-52.21)	
L1.ma			-0.997***	0.985***	-0.997***	
			(-1324.60)	(1164.69)	(-877.71)	
<i>ARCH</i>						
<i>a</i>	0.00950	0.0863	0.0345	0.0165	0.0297	-0.000248
	(0.52)	(1.77)	(0.90)	(0.50)	(1.19)	(-0.01)
<i>g</i>	0.145***	0.501***	0.141**	0.240***	0.0661	0.170***
	(4.47)	(5.09)	(3.11)	(4.36)	(1.72)	(4.42)
<i>d</i>	0.831***	0.246	0.0135	0.198***	-0.356	-0.413**
	(107.91)	(1.95)	(0.04)	(8.29)	(-0.70)	(-2.87)
<i>g + d</i>	0.976	0.747	0.154	0.438	-0.300	-0.243
<i>w</i>	0.270***	2.618***	1.045**	1.406***	0.699**	0.204
	(28.62)	(5.60)	(2.97)	(37.66)	(2.68)	(1.71)
<i>N</i>	5059	5059	5059	5059	5059	5059
<i>AIC</i>	19662.8	23178.2	18127.4	20065.9	15924.0	12748.7
<i>BIC</i>	19715.0	23236.9	18192.7	20124.7	15995.8	12807.5
<i>LBQ(11)</i>	17.130 [0.104]	7.277 [0.776]	18.724 [0.066]	10.798 [0.460]	8.589 [0.659]	11.499 [0.402]
<i>LBQ(25)</i>	32.443 [0.145]	30.882 [0.192]	33.914 [0.109]	35.870 [0.073]	30.052 [0.222]	28.580 [0.281]
<i>LBQ²(11)</i>	14.984 [0.183]	7.333 [0.771]	18.807 [0.064]	10.279 [0.505]	8.402 [0.676]	10.581 [0.479]
<i>LBQ²(25)</i>	30.157 [0.218]	30.687 [0.199]	34.308 [0.101]	33.378 [0.121]	29.985 [0.224]	27.879 [0.313]

t statistics in parentheses and the numbers in square brackets are the *p*-values, *LBQ* is the Ljung-Box statistics for serial correlation and *LBQ²* is the Ljung-statistics for ARCH effect, with the lags used in the parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Given the evidence of the presence of autoregressive conditional heteroschedasticity in the mean of BRICS index returns, we proceed to model the mean and the conditional variance of each of the BRICS market indexes using $ARMA(p,q) - EGARCH(m,n) - M$. The results of our estimates are reported in Table 6 (for daily returns measured in US dollars) and Table 7 (for returns measured in local currency units). The model diagnostic statistics reported at the lower parts of tables 6 and 7 indicates that the estimated mean and conditional variance equations were correctly specified. In particular, the Ljung-Box *Q*-statistic at both lags 11 and 25 did not reject

the null hypothesis that the mean equation is correctly specified for all the BRICS index returns. Also, the Ljung-Box *Q-squared statistic* failed to reject the null hypothesis that the conditional variance equation is correctly specified. The estimated $ARMA(p, q) - EGARCH(m, n) - M$ adequately captures the conditional mean and variance of the return generating process for the BRICS. We therefore turn to the interpretation of the estimated parameters.

According to the results reported in Table 6, the expected daily index return is statistically not different from zero for Brazil, India, China and South Africa. However, the index for Russia and the World stock market index promise a positive return. The estimated mean return for Russia is 0.248 and it is statistically significant at 1% level. For the World index return, the estimated mean is 0.078 which is also statistically significant at 1% error level. When returns are measured in local currency however, the expected daily index returns for Brazil and China are negative and statistically significant at 1% error level. The estimated expected return for Brazil and China are -0.121 and -0.084 respectively. On the contrary, Russia and the world index returns are both positive and statistically significant on the average, when returns are measured in local currency units. For Russia and the World, the estimated expected daily index returns are 0.197 (which is statistically significant at 5% error level) and 0.103 (which is statistically significant at 1% error level) respectively. The estimated expected daily index returns in local currency units are, however, not statistically different from zero for India and South Africa. Thus, for India and South Africa the daily index return is zero and positive for Russia, irrespective of the unit in which returns are measured.

One implication of the capital asset pricing model (CAPM) is that expected return and risk are related positively. For a rational risk-averse investor to take on an additional risk, he must be compensated with additional return. We confront this prediction of CAPM with BRICS index returns data. The risk-return relationship parameter is captured by the coefficient on *sigma-squared* of the ARCHM part of the results reported in Tables 6 and 7. From Table 6 the estimated coefficient on the risk variable (appropriately interpreted as the *risk premium*) is positive for all the BRICS except Russia and the World, for daily dollar returns. In sharp contrast with the predictions of the CAPM, the coefficient on the conditional variance in the mean equation which captures the relationship between risk and return are negative and statistically significant for Russia and the World. This is a violation of rational behaviour of risk-averse investors in equity

markets. However, the estimated risk-return relationship is only statistically significant for Russia, China (10% level of significance) and the World (5% level of significance). With returns measured in local currency however, the estimated coefficient on the conditional variance in the mean equation is positive and statistically significant for Brazil and China but negative and significant for the World. However, given the negative mean local currency return for Brazil and China, the estimated positive relative risk aversion coefficients implies a negative relationship between risk and return, a finding which is inconsistent with CAPM. The World local currency index return also violates the predicted positive risk-return relationship by CAPM since the estimated mean return is positive while the estimated risk aversion parameter is negative and statistically significant.

We now turn to the question of predictability of daily index returns both in US dollars and local currency units and the implications thereof for the efficiency of BRICS markets. In well-functioning equity markets the prices of the securities traded act as though they fully reflect all available information and react instantaneously and in an unbiased fashion to new information [Fama, 1970, 1991]. If this were not the case, it would be possible to obtain riskless return, casting doubts about the ability of the stock market to efficiently allocate capital. The weak form of the efficient markets hypothesis thus precludes any predictability on the basis of past information (see Fama, 1991). Are BRICS stock returns predictable?

This question can be answered from the $ARMA(p,q) - EGARCH(m,n) - M$ component of the estimated $ARMA(p,q) - EGARCH(m,n) - M$ reported in Tables 6 and 7. According to the results reported in Table 6, the coefficient on the AR(1) is positive and statistically significant at 1% error level for Brazil, India, China and the World daily returns. Not only are these coefficients statistically significant, but also economically given the large sizes of the coefficients. For instance, the coefficient AR(1) are 0.999, 0.990, 0.998 and 0.057 for Brazil, India, China and the World respectively. Interestingly these sub group for the BRICS markets also showed negative and statistically significant coefficient on the first lag of the moving average (MA) component of the estimated $ARMA(p,q) - EGARCH(m,n) - M$. The estimated coefficient of the first lag of the shock to the mean process is -0.997 for Brazil, -0.997 for India and -0.997 for China. The implication from this is that past information about returns (prices) have predictive power on current and future returns (prices), a sharp contrast to the weak-form efficiency hypothesis. In the case of Russia and South Africa, there is no evidence that past returns and shocks to return

prove any information in predicting current and future returns (prices), which is in consonance with the weak-form efficiency hypothesis.

Measuring returns using local currency series changes the results for Brazil and South Africa. In the case of Brazil neither past returns nor shocks to it offer any predictive power on current and future returns. Thus, the results from returns measured in local currency units for Brazil are consistent with the weak form efficiency hypothesis. This is in sharp contrast with the finding obtained when we measure returns in US dollars. For South Africa there is positive and statistically significant AR (1) and a negative and statistically significant MA(1) effect in the return generating process. This implies that past returns and past shocks to returns can be used to predict current and future daily Rand returns in the South African stock market. This finding is a violation of the weak form efficiency hypothesis of capital markets. The evidence for China, India and the World, as far as their implication for efficiency is concerned is invariant to whether returns are measured in local currency units or US dollars. On the whole, there is conclusive evidence that the stock markets of China, India and the World are not efficient while that of Russia is at least weak form efficient. Efficiency implications for Brazil and South Africa are sensitive to units in which returns are measured. The efficiency of otherwise of the forex markets in these economies has some contagion effects of the efficient functional of their capital markets. The predictability of BRICS returns seem to concur with similar studies on emerging markets [see for example Cooper, 1982; Darrat and Mukherjee, 1986; Errunza and Losq, 1985], which examined the weak form version of the efficient markets hypothesis. The presence of first order first-order serial correlation in stock prices shows that information may not be fully incorporated in security prices. Dailami and Atkin (1990) argue that positive serial correlation may result in slow incorporation of new information, insider trading, or infrequent trading. There may be barriers to the dissemination of information, and companies appear to divulge less information with a greater time lag than is the norm. On the other hand, negative serial correlations may signal thin trading and subject to speculative influences.

4.1. Volatility persistence

Volatility persistence in the return series for each of the countries is described by both the L_{earch_a} and L_{egarch} terms in Tables 6 and 7. The L_{earch_a} denotes the lagged squared residuals from the EGARCH-M model, while the L_{egarch} reflects the lagged conditional variance term in

the EGARCH model. Volatility is said to be persistent if the sum of the two volatility terms is close to unity, less persistent if less than unity and explosive if greater than unity. In both return series (in dollars and local currency) as reported in Table 6 and 7, we find strong indication of volatility persistence for Brazil, the magnitude of the persistence terms sum-up to close to unity (0.944 and 0.976 for returns in dollars and local currency, respectively). We find less volatility persistence in the return series for all the other countries and also the World return, irrespective of the unit of currency (dollars or local currency). The implication of these results is that, all the BRICS countries (with the exception of Brazil) show no evidence of long-memory in their respective return series. This means that shocks in volatility to returns of most of the BRICS countries and that of the World tends to decay very quickly, implying that previous volatility do not have a strong predictive power on current volatility, except in the case of Brazil. These results are, however, conditional on the model specification and the distribution assumption made in the estimation. Since all descriptive statistics and normality test along with testing for the effect of autoregressive conditional heteroskedasticity pointed towards the model used in estimating the parameters presented in Tables 6 and 7, these results are very satisfactory.

4.2. Asymmetry in return

An avalanche of empirical research in emerging market returns distribution point to significant leverage effect, where higher volatility tend to follow negative returns(see Alagidede, 2011 and references therein). Asymmetries in the distribution of returns may arise either because of shocks to systematic risk factors that affect the cross section of returns, or because of country-specific shocks. This is taken up in the results reported in Tables 6 and 7. The evidence reveals a positive leverage effect for all return series (both in dollars and local currency), except that for the World , which has a negative leverage effect, implying that negative news has greater impact on volatility than positive news. However, the estimated leverage effect coefficient for each of the return series are not significantly different from zero at the 5% level suggesting the absence of leverage effects in these markets. The only market with a significant positive leverage effect is the Russian market and this is only true at the 10% significant level. The conclusion one can make in regards to asymmetry based on these results is that, evidence tend to support symmetric effects of news or shocks for each of the return series, however, the Ling-McAleer (2000) and McAleer et al. (2007) test (results reported in the appendix) to discriminate between symmetric GARCH-M and the asymmetric EGARCH-M models, in majority of the cases supported the EGARCH-M as the best model. With the exception of the weak evidence in Russia we

collaborate the findings of Bekaert and Harvey (1995) and Bekaert and Harvey (1997) who do not find support for leverage effects in emerging markets. The implication is that irrespective of not finding significant leverage effects in statistical terms it appear to be important to incorporate such effects in the modelling strategy to reflect the data generating process.

5. Conclusion

Stock returns in emerging market economies exhibit patterns that are distinctively different from developed economies. Previous research has established that standard asset pricing models consistently fail to account for all the peculiarities of emerging markets returns. For instance, returns are noted to be highly volatile and non-normally distributed. Long horizon returns are predictable while there is significant autocorrelation in returns. After over two decades of research in emerging market economies, the issue of return distribution is far from settled. New data and empirical techniques, coupled with faster growth and increasing importance of emerging markets allow us to re-examine the distribution of stock returns for Brazil, Russia, India, China and South Africa. In particular, the validity of the weak form version of the efficient markets hypothesis is tested, and stylized facts of volatility clustering asymmetry, long memory, and leverage effect for our sample is examined for the period 1995 to 2014.

Using a multivariate test for return distribution, and accounting for risk *premia* and conditional volatility we arrived at the following findings: a) the distribution of stock returns for the BRICS exhibits peakedness with fatter and longer tails. This is invariant to both the unit of measurement and the time horizon in which returns are studied. We argue that the underlying structure of the BRICS, particularly their potential for large growth swings, susceptibility to regulatory and political changes may lead their stock returns to deviate significantly from the Gaussian assumption and this ought to be incorporated in any inferences on return distribution. b) While the stock markets of China and India are predictable irrespective of unit of measurement, return predictability for Brazil and South Africa are conditional on whether we are looking at local currency or US dollar returns. C) All markets exhibit volatility clustering, and while this decays for most markets, it tends to be persistent for Brazil. c) The so called leverage effect is found to be weak in our sample.

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Appendix A: Summary Statistics of BRICS Returns

Daily US Dollar Returns						
	BRAZIL	RUSSIA	INDIA	CHINA	SOUTH AFRICA	WORLD
Mean	0.023	0.039	0.024	-0.003	0.020	0.020
Median	0.058	0.060	0.021	0.003	0.076	0.068
Maximum	17.335	24.220	19.486	14.044	12.353	9.097
Minimum	-18.323	-31.013	-12.041	-14.442	-13.566	-7.325
Std. Dev.	2.318	2.992	1.738	1.971	1.726	0.993
Skewness	-0.097	-0.447	-0.053	0.029	-0.419	-0.371
Kurtosis	10.585	14.836	9.636	8.772	8.482	10.819
Jarque-Bera	12135.07 [0.000]	29699.48 [0.000]	9284.83 [0.000]	7024.63 [0.000]	6483.65 [0.000]	13002.46 [0.000]
Daily Local Currency Returns						
Mean	0.042	0.040	0.036	-0.003	0.038	0.019
Median	0.000	0.061	0.000	0.000	0.013	0.068
Maximum	24.734	24.220	16.423	14.036	6.750	8.720
Minimum	-14.217	-31.013	-12.050	-14.457	-12.208	-7.156
Std. Dev.	1.923	2.928	1.588	1.969	1.253	0.960
Skewness	0.351	-0.462	-0.133	0.025	-0.406	-0.337
Kurtosis	15.365	15.834	9.392	8.811	8.249	10.360
Jarque-Bera	32334.44 [0.000]	34900.44 [0.000]	8626.06 [0.000]	7118.92 [0.000]	5947.81 [0.000]	11515.55 [0.000]

Weekly US Dollar Returns						
	BRAZIL	RUSSIA	INDIA	CHINA	SOUTH AFRICA	WORLD
Mean	0.125	0.196	0.124	-0.011	0.099	0.101
Median	0.422	0.447	0.364	0.234	0.356	0.321
Maximum	25.617	44.899	18.366	22.536	27.601	11.636
Minimum	-33.056	-31.698	-21.879	-24.336	-18.855	-22.381
Std. Dev.	5.322	7.100	3.950	4.619	3.907	2.375
Skewness	-0.650	-0.125	-0.369	-0.269	-0.193	-1.135
Kurtosis	7.075	8.584	5.247	5.956	7.666	12.853
Jarque-Bera	770.805 [0.000]	1316.244 [0.000]	235.577 [0.000]	380.276 [0.000]	923.208 [0.000]	4306.687 [0.000]
Weekly Local Currency Returns						
Mean	0.221	0.200	0.185	-0.011	0.191	0.096
Median	0.440	0.448	0.434	0.230	0.296	0.323
Maximum	19.189	44.916	13.660	22.422	16.264	10.736
Minimum	-22.529	-31.698	-19.000	-24.341	-13.454	-21.318
Std. Dev.	4.120	6.946	3.557	4.609	2.840	2.287

Skewness	-0.547	-0.104	-0.332	-0.276	-0.142	-1.040
Kurtosis	6.872	9.157	5.276	5.989	6.107	12.208
Jarque-Bera	681.831 [0.000]	1598.525 [0.000]	236.676 [0.000]	389.233 [0.000]	409.969 [0.000]	3753.445 [0.000]

Monthly US Dollar Returns						
	BRAZIL	RUSSIA	INDIA	CHINA	SOUTH AFRICA	WORLD
Mean	0.552	0.945	0.564	-0.008	0.478	0.451
Median	2.027	3.095	1.397	1.240	1.263	1.080
Maximum	24.042	42.580	28.276	30.669	24.838	11.916
Minimum	-61.139	-72.397	-43.462	-36.550	-54.335	-31.643
Std. Dev.	10.929	16.065	9.314	10.156	8.513	5.232
Skewness	-1.140	-1.056	-0.651	-0.539	-1.470	-1.676
Kurtosis	6.915	7.011	5.177	4.109	10.021	9.790
Jarque-Bera	198.365 [0.000]	198.685 [0.000]	62.238 [0.000]	23.136 [0.000]	560.055 [0.000]	554.251 [0.000]
Monthly Local Currency Returns						
Mean	0.967	0.961	0.832	-0.007	0.867	0.437
Median	1.651	2.715	1.395	1.250	1.370	1.237
Maximum	20.928	42.580	26.534	30.705	16.065	13.365
Minimum	-38.265	-71.604	-34.906	-36.677	-23.775	-26.982
Std. Dev.	8.059	15.652	8.315	10.147	6.004	4.987
Skewness	-0.864	-1.097	-0.534	-0.545	-0.761	-1.516
Kurtosis	5.709	7.432	4.802	4.135	5.086	8.405
Jarque-Bera	99.811 [0.000]	236.415 [0.000]	42.394 [0.000]	23.922 [0.000]	64.468 [0.000]	371.243 [0.000]

Appendix B: Test for Non-nested models

The Ling and McAleer (2000) and McAleer et al.(2007) test is a testing procedure for non-nested models, for instance between EGARCH and GARCH models. Suppose our proposed model is the EGARCH and we want to compare that with the GJR-GARCH model, the proposed testing procedure is to check for significant coefficient of the log variance from the GJR-GARCH model in the following specification:

$$\ln(h_t) = \omega + \alpha |z_{t-1}| + \gamma z_{t-1} + \beta \ln(h_{t-1}) + \delta \ln(\hat{\sigma}_t^2)$$

Where the estimated variance from the EGARCH model is h_t , $\hat{\sigma}_t^2$ is the estimated variance from the GJR-GARCH model and z_{t-1} is the standardized residuals from the EGARCH model. The Null hypothesis is that; $\delta = 0$, which is in effect testing for significance of the δ in the above equation and non-significance implies $\delta = 0$ and therefore the EGARCH in this case will be the preferred model relative to the GJR-GARCH model.

Rejection percentage of the Null hypothesis of $d'=0$ base on Ling-McAleer (2000) test

Test at 5% sig. level	Null Model	Alternative	Local Currency	Dollars
Ling-McAleer	EGARCH	GJR-GARCH	0%	0%
Ling-McAleer	EGARCH	GARCH	0%	33%